

SSD2353 HIGH PERFORMANCE AIOT SYSTEM-ON-CHIP

PRODUCT BRIEF

CHIP OVERVIEW

The SSD2353 is a highly integrated System-on-Chip (SoC) dedicatedly designed for personal mobile device, battery device, and AIoT application.

The SSD2353 includes a 64-bit quad-core processor, advanced Image Signal Processor (ISP), Intelligence Processing Unit (IPU), Intelligence Video Engine (IVE) as well as high speed I/O interfaces like USB, Ethernet, and 12-bit ADC. These features in combination make the SSD2353 an ideal solution that facilitates design and development of high-performance, high-picture-quality, and low-cost products.

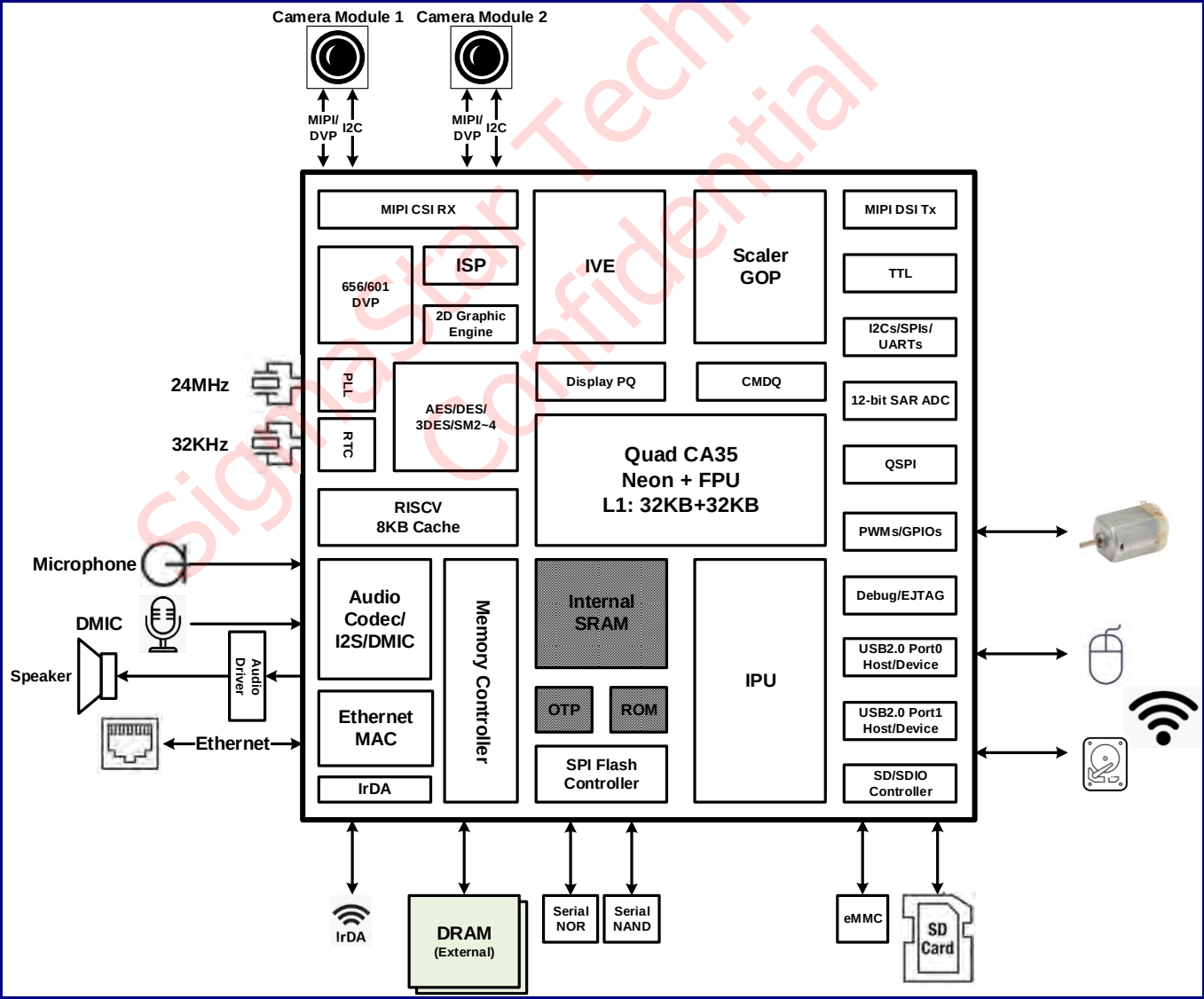
The programmable neural network inference engine featured in the SSD2353 allows customers to achieve a rich variety of intelligent applications with ease.

Implemented with the quad-core ARM Cortex-A35 CPU as well as an IPU, the SSD2353 enables fast startup, real-time performance, and connections with various peripheral interfaces.

Efficient computing resources are available to help customers develop industry and consumer applications. Advanced low-power, low-voltage architecture and optimized design flow are implemented to fulfill long time usage applications. Hardwired AES/DES/3DES as well as SM2/SM3/SM4 cipher engines are integrated to support secure boot, authentication, and video/audio stream encryption in security system.

The SSD2353, powered by SigmaStar Technology, comes with a complete hardware platform and software SDK, allowing customers to speed up "Time-to-Market."

BLOCK DIAGRAM



FEATURES

■ High Performance Processor Core

- ARM Cortex-A35 Quad Core
- 32KB L1 I-cache and 32KB L1 D-cache for each core
- 256KB L2 cache
- Neon and FPU
- Separate power domain for each core
- Stand-alone voltage domain

■ Video Input Interface

- Supports 8/10 parallel interface for raw data input
- Supports 10-bit BT601 interface
- Supports 8-bit BT656 interface
- Supports MIPI interface with 2 data lanes and 2 clock lanes
- Supports sensor interface with both parallel and MIPI
- Static and adaptive bad pixel compensation
- Crosstalk noise reduction
- Temporal-domain Noise Reduction (3DNR)
- Sharpening filters for image enhancement
- Spatial-domain Noise Reduction (2DNR) for luma and chroma image
- Filter to remove purple false color in highlight regions
- Symmetric/Asymmetric lens shading compensation
- Auto White Balance (AWB) / Auto Exposure (AE) / Auto Focus (AF)
- CFA color interpolation and demoiré filter
- Color correction and color adjustment engine
- Gamma correction
- Frame buffer data compression and de-compression to save memory bandwidth
- Wide Dynamic Range (WDR) with local tone mapping
- Fully programmable multi-function scaling engines

■ Intelligence Video Engine (IVE)

- Pure hardwired accelerator
- Supports image process operator like Filter2D, Box filter, Gaussian filter, Bernsen, Dilate, Erode, etc.
- Supports Matrix multiplication

■ Intelligence Processing Unit (IPU)

- Pure hardwired accelerator
- Programmable 4/8/16-bit process
- Supports RGB/YUV data format R/W DMA
- Supports various video analysis functions like FD/FR, human detection, MD/OD, object tracking, etc.
- Supports Transformer networks

■ Co-Processor (RISCV)

- Supports RV32 base instruction set, and M/C extension
- 6 Stage Pipeline, Single Issue, in-order dispatch, out-of-order execution
- 8Kbyte + 8Kbyte configurable I/D cache or I/D TCM
- Up to 576MHz clock rate
- Supports dynamic branch prediction
- Supports memory property configuration
- Supports JTAG debug

■ Audio Processor

- Supports 3-channel ADC with single-end or differential mode
- Supports 2-channel DAC with single-end mode
- ADC and DAC SNR over 95.2dB
- Digital and analog gain adjustment
- Supports 8-CH DMIC (1 clock + 4 data)
- Supports I2S0 TDM mode with input max. 8-ch and output 2-ch
- Supports I2S1 2-ch input and 2-ch output
- Supports I2S2 2-ch input and 2-ch output
- I2S0/1/2 support Master or Slave mode and 4/6 wire mode, bit width up to 32-bit
- Supports SPDIF input, bit width up to 24-bit

■ Video Output Interface

- Picture quality enhancement (gamma, sharpness, brightness, 3x3 matrix, color dither, osdb, RGB swap)
- Display channel can output to MIPI/Digital port, digital port including one of TTL/CCIR656/18080
- Supports MIPI DSI TX 4lane, 2.5Gbps per lane, RGB 16/18/24-bit, 2560x1600@60fps
- TTL/Parallel-RGB interface, 16/18/24-bit, 1280x800@60fps
- Supports 8/16-bit 18080 interface

■ Advanced Color Engine

- Luma gain/offset adjustment
- Black/White Level Extension (BLE/WLE)
- Peaking/LPF/FCC/Coring

■ SPI NOR/NAND Flash Interface

- Compliant with standard, dual and quad SPI flash memory components
- Max. 108MHz clock rate
- Power failure protection

■ SD/eMMC Interface

- Compatible with SD spec. 2.0, data bus 1/4-bit mode
- Compatible with SDIO spec. 2.0, data bus 1/4-bit mode
- eMMC 5.0 with 4/8 data bit and max. 200MHz clock rate, HS400 DDR mode

■ USB Interface

- USB2.0 port0 configurable Host or Device
- USB2.0 port1 configurable Host or Device (Host mode supports EHCI specifications)

■ DRAM Memory

- External 16-bit x1 DDR4
- Data rate up to 2666Mbps for DDR4

■ Connectivity

- Two built-in 10/100M Ethernet MACs with RMII (EMAC x2)
- Hash table with 256 entries
- Broadcast/Multicast storm prevention
- Supports both full-duplex and half-duplex operation
- Supports IEEE 802.1Q VLAN tag detection for reception frames
- Supports checking IPv4 header checksum and TCP, UDP, or ICMP checksum encapsulated in IPv4 or IPv6 datagram

■ Security Engines

- Supports AES128/AES192/AES256/DES/3DES/RSA4096/SHA-1/SHA-256/SM2/SM3/SM4
- Supports secure booting
- FIPS 140-1 compliant random number generator
- Embedded OTP (One Time Programmable) memory to store secure and calibration data

■ Boot Options

- SPI NOR
- SPI NAND with ECC
- SD Card
- eMMC
- USB
- UART

■ Peripherals

- Dedicated GPIOs for system control
- Supports 8x PWM inputs and 20x PWM outputs (shared with GPIOs)
- Supports PWM in capture which contains orthogonal encoder mode (AB phase calculation), pulse/direction count calculation, etc.
- Supports QSPI interface
- Up to 7 generic UARTs and 4 fast UARTs with flow control
- Up to 12 generic timers and one watchdog timer
- Two SPI interfaces, which can be configured as master or slave mode
- Two SPI interface, which is master mode only
- Up to six I2C Masters
- Built-in 10-bit SAR ADC with max. 5-channel analog inputs for different kinds of application
- Built-in 12-bit SAR ADC (up to 1.5Msps) with max. 22-channel analog inputs for different kinds of application
- Built-in 12-bit SAR ADC (up to 5Msps) with max. 2-channel analog inputs for different kinds of application
- Supports 7x7 Keypad
- Supports IrDA
- Supports POR (Power On Reset)
- Supports internal temperature sensor

■ Real Time Clock (RTC)

- Built-in RTC working with 32.768 KHz crystal
- Alarm interrupt for wakeup
- Tick time interrupt (millisecond)
- Built-in regulator
- Supports low leakage RTC mode for long battery application

■ Always on power domain (PM)

- Built-in RC FRO to generate clock source
- Supports internal co-processor to control PM GPIO
- Supports multiple GPIOs for power control and RTC events

■ Package

- BGA 14x14
- Ball pitch and size: 0.65 and 0.3 mm